CS 210: Worksheet 13 *Dr. Mudassir Shabbir, October 13, 2026*

Student ID 1: _____ **Seat Location 1:** row:_____ col:_____

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Cartesian products, relations, and their properties. Begin with computations and finish with proofs and constructions. Show your reasoning. Prove claims or give checked counterexamples.

1. *Easy*

Let $A = \{a\}$, $B = \{b\}$, and $C = \{c\}$. Treat triples and nested ordered pairs as different kinds of objects.

Write the sole element of $A \times B \times C$, of $(A \times B) \times C$, and of $A \times (B \times C)$.

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2. *Easy*

Can an ordered pair be built using only sets? Define $K(a, b) = \{\{a\}, \{a, b\}\}$.

Compute $K(1, 2)$, $K(2, 1)$, and $K(1, 1)$.

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3. *Moderate*

Let $A = \{0, 1, 2, 3, 4\}$ and $B = \{0, 1, 2, 3\}$. Define relations from A to B by $aRb \iff a + b = 4$ and $aSb \iff a > b$.

Find $R \cap S$ and R^{-1}. State the source and target sets of R^{-1}.

4. *Moderate*

On $A = \{1, 2, 3, 4, 5, 6\}$, let aDb mean that a divides b.

(a) Construct its 6×6 zero-one matrix, using the order $1, 2, 3, 4, 5, 6$.

(b) Decide whether D is reflexive, symmetric, antisymmetric, and transitive. Justify each answer.

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5. *Moderate*

Let A have $n \geq 2$ elements, and fix distinct $a, b \in A$. Explain each count by identifying the freely selectable pairs.

(a) How many relations on A are there? How many must contain (a, b)?

(b) How many contain no pair whose first coordinate is a? How many contain at least one such pair?

6. *Moderate*

Let A, B, C, D be sets.

(a) Prove that $A \subseteq C$ and $B \subseteq D$ imply $A \times B \subseteq C \times D$.

(b) If A, B are nonempty, prove the converse: $A \times B \subseteq C \times D$ implies $A \subseteq C$ and $B \subseteq D$.

7. *Challenge*

Symmetry and antisymmetry are not opposites.

(a) Give a relation on $\{1, 2\}$ that is both symmetric and antisymmetric.

(b) Construct a relation that is neither. What is the smallest possible number of underlying elements? Prove that a smaller set cannot work.

Relation properties: quick reference. For $R \subseteq A \times A$, write aRb for $(a, b) \in R$. All statements below hold for every $a, b, c \in A$.

Reflexive: every element relates to itself.
aRa.

Irreflexive: no element relates to itself.
$\neg(aRa)$.

Symmetric: every pair has its reverse.
$aRb \Rightarrow bRa$.

Antisymmetric: mutual pairs force equality.
$(aRb \wedge bRa) \Rightarrow a = b$.

Asymmetric: no pair has its reverse.
$aRb \Rightarrow \neg(bRa)$.

Transitive: every two-step chain has a shortcut.
$(aRb \wedge bRc) \Rightarrow aRc$.

Equivalence relation: reflexive, symmetric, and transitive.